

PROJECT TITLE: LearnSphere AI – An AI-Powered EdTech Platform

BACKEND TECHNOLOGY STACK

- Framework: FastAPI
- Programming Language: Python 3.11+
- Authentication: JWT (JSON Web Token) with Role-Based Access Control
- AI Provider: Groq AI (AI Tutor and Mock Interview Evaluation)
- Local MCP Integration: Restricted local knowledge responses
- Certificate/PDF Generation: PDF generation library
- API Documentation: OpenAPI and Swagger UI
- Server: Uvicorn (ASGI)
- Package Manager: NPM (frontend) and Python dependency management (backend)

BACKEND MODULES

Authentication Service

- User registration and login
- Password recovery and reset
- JWT token generation and validation
- Student, instructor, and administrator role checks

User and Profile Service

- Current-user profile retrieval and update
- Password change
- Role-aware dashboard data
- Enrollment and lesson-progress history

Course Service

- Course creation and listing
- Search and course details
- Instructor ownership checks
- Free and paid course support

Lesson and Upload Service

- Lesson creation and ordering
- PDF, video, note, and resource content
- Protected content download
- Lesson completion tracking

Enrollment Service

- Free-course enrollment
- Student enrollment status
- Access control for protected learning content

Assessment Service

- Assessment creation
- Question and option management
- Quiz submission and scoring
- Attempt and answer review

Wallet and Payment Service

- Wallet balance retrieval
- Wallet top-up
- Paid-course purchase
- Transaction history

AI Tutor Service

- General and course-aware questions
- Groq AI responses
- Restricted local MCP responses
- Chat history persistence

Analytics Service

- Student course progress
- Instructor course performance
- Enrollment, lesson, attempt, and average-score metrics

Certificate Service

- Course-completion eligibility
- Certificate generation
- Certificate listing and PDF access

Recommendation Service

- Suggested courses not yet enrolled
- Popularity and enrollment-based reasons

Study Planner Service

- Create study plans
- Target dates and daily minutes
- Complete and delete plans

Mock Interview Service

- Interview question bank
- AI answer evaluation
- Scores, feedback, and attempt history

Notification Service

- User notification list
- Read and unread state
- Certificate and platform updates

Instructor KYC Service

- Identity-document upload
- Submission status
- Administrator approval or rejection

Administration Service

- Platform statistics
- User role management
- KYC review
- Course monitoring and deletion

BFF Dashboard Service

- Student dashboard aggregation
- Instructor dashboard aggregation
- Administrator dashboard aggregation

DATABASE TABLES

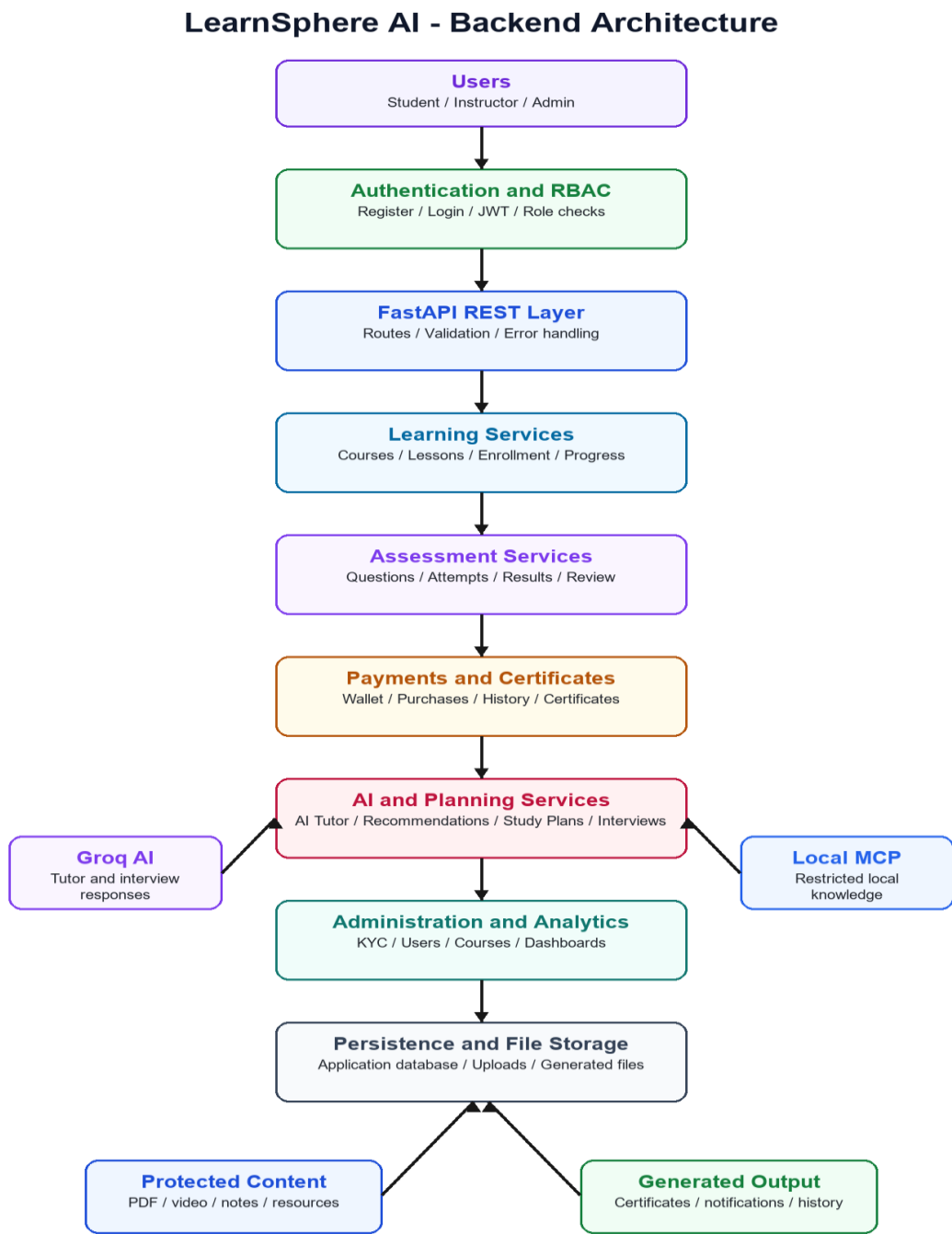
- Users
- Courses and Enrollments
- Lessons and Lesson Progress
- Assessments, Questions, Attempts, and Attempt Answers
- Wallet and Payments
- Certificates
- Notifications
- Study Plans
- Interview Questions and Interview Results
- AI Chat History

KEY FEATURES

- Secure role-based student, instructor, and administrator access
- Free enrollment and wallet-based paid-course purchasing
- Protected lesson content with completion tracking
- Assessment authoring, quiz attempts, automatic scoring, and answer review
- AI Tutor with course context and saved chat history
- Learning progress, instructor analytics, and certificate generation
- Course recommendations, study planning, and AI mock interviews
- Instructor KYC verification and administrator review workflows
- Responsive AppLayout with shared sidebar, top bar, and reusable PageHeader

ARCHITECTURE APPROACH

The backend follows a modular service-oriented architecture. FastAPI acts as the central REST layer and coordinates authentication, courses, lessons, assessments, payments, AI assistance, analytics, certificates, planning, interviews, notifications, and administration. Authorization and resource ownership are enforced by the backend for every protected operation.



FRONTEND TECHNOLOGY STACK

- Framework: React with TypeScript
- Build Tool: Vite
- UI Library: Material UI and Material Icons
- Routing: React Router
- HTTP Client: Shared Axios-style API client
- AI Response Rendering: React Markdown with remark-gfm
- Package Manager: NPM
- Development Environment: VS Code

FRONTEND MODULES

Authentication Module

- Registration and login
- Forgot and reset password
- Protected-route handling

AppLayout Module

- Permanent role-based sidebar
- Shared top bar
- Responsive navigation
- Reusable PageHeader

Dashboard Module

- Role-specific summaries
- Quick links to permitted features
- Recent notifications

Course Discovery Module

- Course search and course cards
- Course details with free and paid labels

Student Learning Module

- My Courses
- Lesson content and completion status
- Course assessments

Assessment Module

- Quiz questions and required answers
- Results and answer review

Payment Module

- Wallet balance and top-up
- Course purchase and transaction history

AI Tutor Module

- Course context selection
- Markdown responses with source labels
- Chat history

Progress and Certificate Module

- Progress bars and lesson/assessment metrics
- Certificate generation and PDF view

Planning and Interview Module

- Study plans
- Mock interview questions
- AI feedback and interview history

Instructor Module

- KYC verification
- Course creation and lesson upload
- Assessment management and analytics

Administrator Module

- Users and roles
- KYC decisions
- Course monitoring and question bank

Profile and Notification Module

- Profile update and password change
- Read and unread notifications

FRONTEND FEATURES

- Responsive user interface with modern dashboard design
- Secure authentication flow with JWT handling
- REST API integration with FastAPI backend via shared Axios client
- Interactive learning progress visualizations
- AI-powered chat experience with markdown rendering
- Role-specific navigation and access control
- Protected content download and certificate PDF access
- User-friendly course discovery and enrollment flow

FRONTEND ARCHITECTURE

The frontend uses route-based React pages inside a protected `AppLayout`. Each page owns its domain state, loading state, alerts, and API calls. `AppLayout` owns permanent application navigation, while `PageHeader` owns the page title, subtitle, contextual back route, and optional actions. This removes duplicate `AppBar` and `Toolbar` navigation from individual pages.

LearnSphere AI - Frontend Flow Diagram

